Worksheet — Horizontal Launch Simulation

Use with *Simulation - Horizontal launch.html* · A.1 Kinematics: projectile launched horizontally, horizontal and vertical motion treated separately; goes with the ball-off-the-bench experiment (carbon paper)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/11-horizontal-launch** in a browser (offline: *Simulation - Horizontal launch.html*). Set the height h and the launch speed u, then ▶ **Launch**. Ball B is dropped from the edge at the same moment. The strobe shows the ball every 0.1 s and the shadows split the motion into horizontal (floor) and vertical (wall) parts.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 15 min) · **Part 3** homework (≈ 30 min).

Use $g = 9.81$ m s^{-2}. The two directions share only the time t: horizontal $a_x = 0$, $x = ut$; vertical $a_y = -g$, from $v_y = 0$.

Scan to open

Part 1 — Pre-lab: predict, then test

Write your prediction **before** you run the simulation. Then test it and explain what you saw.

Question	My prediction (before)	What the simulation showed	Physics explanation
1. Ball A rolls off the edge at 2.0 m s^{-1}; ball B is dropped from the same height at the same moment. Which lands first?			
2. The launch speed is doubled (same height). What happens to the time of flight and to the range?			
3. What happens to the horizontal velocity v_x while the ball is in the air?			
4. What shape is the path? Sketch it.			

5. Your bench is 0.90 m high. Calculate the time of flight of a ball rolling off it. Then check with the simulation.

Part 2 — During the lab: compare with the experiment

Class experiment. Roll the ball down the ramp, off the bench, onto carbon paper. Measure the bench height h and the range R from the edge (use a plumb line) for 3 launches.

h / m	R_1 / m	R_2 / m	R_3 / m	mean R / m	$t = \sqrt{(2h/g)}$ / s	$u = R/t$ / m s^{-1}

Simulation. h = 0.90 m, u = 1.5 m s^{-1}, landing mark = carbon paper. Launch 3 times and ● **Record**.

h / m	R_1 / m	R_2 / m	R_3 / m	mean R / m
0.90				

1. Mean range from the simulation: R = _____ m

2. A group measured R = 0.64 m from the 0.90 m bench. Calculate the launch speed of their ball.

3. Group 1's ball lands twice as far from the bench as group 2's (same bench). What can you conclude? Explain.

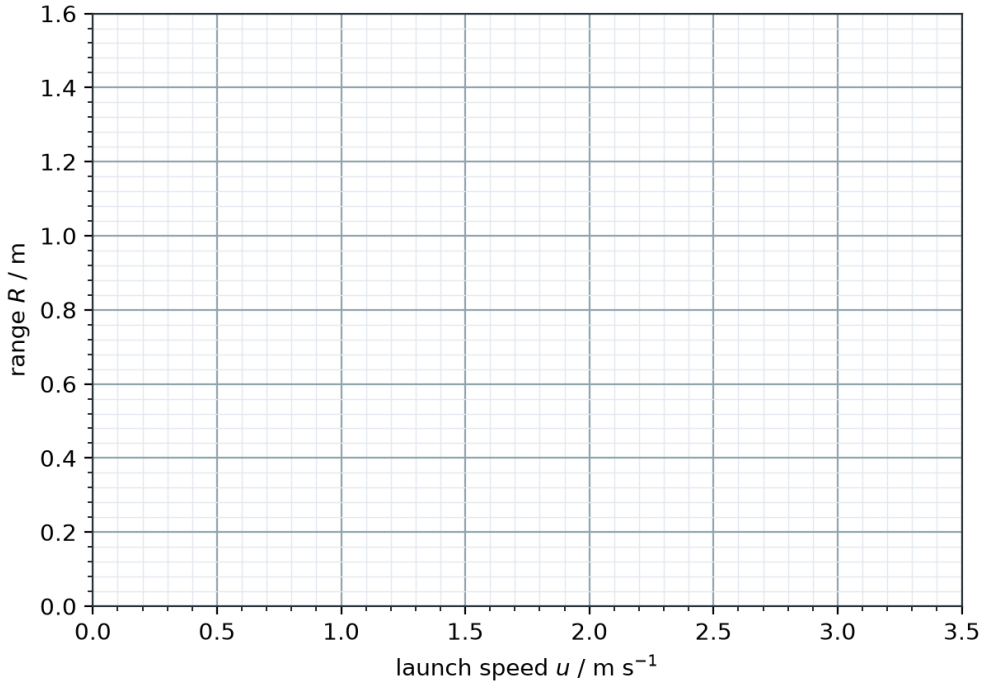
4. Give two sources of uncertainty in the class experiment and say how to reduce each.

Part 3 — Homework: analyse

h = 0.90 m, landing mark = carbon paper. Record one launch at each speed.

u / m s^{-1}	R / m
0.5	
1.0	
1.5	
2.0	
2.5	
3.0	

1–2. Plot R (y-axis) against u (x-axis). Draw the best-fit line through the origin.



3. Gradient = _____ s. What does the gradient represent? Use it to find g.

4. A ball rolls off a 1.20 m high table at 3.0 m s^{-1}. How far from the foot of the table does it land?

5. Calculate the speed of the ball just before it lands.

6. At what angle below the horizontal does it land? Draw the velocity triangle.

7. A stone is thrown horizontally from a 20 m high cliff and lands 30 m from the foot of the cliff. Calculate its launch speed.

8. The ball of Q4 is launched in the same way on the Moon ($g = 1.62\ m\ s^{-2}$). Calculate the range and compare.